Data sources for data review
Supporting Document `IndigoCarbon_US-1_2024_0069`

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1 Agricultural Resource Management Survey (ARMS)

The Agricultural Resource Management Survey (ARMS) is a yearly survey of US farmers "on the production practices, resource use, and economic well-being of America's farms and ranches" (see ARMS home). The survey is sponsored jointly by the USDA's Economic Research Service (ERS) and National Agricultural Statistics Service (NASS), and they release two different data products:

- Farm structure and finance, and

- Crop production practices.

Because the survey is annual and reports are released slowly (the latest data are from 2015), we retrieved the data manually by downloading a set of CSVs, then merged these data into a single dataset that was imported into UDP, Indigo's centralized data warehouse. The CSVs were downloaded in May 2020.

ARMS crop practices data are aggregated at the state and national level, and sometimes at the regional level. (See the ARMS III Farm Production Regions Map of "ARMS regions".) For each geography–year pair, we get an estimated value, an error measure, and a flag indicating low sample size for a relatively large set of variables. Because the expected use pattern is to find the latest value for a given geography and variable, and the data are basically read-only (i.e., rarely updated and always as a batch by data owners), and we made the following design decisions.

- The data are stored in a single table, rather than normalized to a set of related tables. That makes most user queries very simple, and because the data are static, it is enough to enforce during data prep and import.
- The table is organized to have one record per geography–year–variable triple, rather than one very wide record per geography–year. In other words, it is a long table rather than a wide table. This format better matches our expected queries and requires less transformation from the original CSVs (and thus less chance for mistakes).

The table schema is below.

Column Name	Type	Description
id	Integer, primary key	Primary key
report	String, not null	The source data are grouped into a set of separate "reports," (e.g., Crop Residue Management Practices or Manure Table). The report name is needed to help interpret some of the variable names, which can be rather obscure. This column will be null for any variable that appears in multiple tables (e.g., Planted Acres).
year	Integer, not null	Year data were collected
state	String	Full state name for data aggregated to state level. This column is null for national-level data.
state_abbr	String	Two-letter state abbreviation
state_fips	Integer	State FIPS code
crop	String, not null	Crop name, standardized within this table, but not matched to any Indigo-wide crop name table.
variable	String, not null	Name of the measured variable
units	String, not null	Units of the measured variable
estimate	Float	Best-estimate variable value
relative_std_err	Float	Relative standard error for variable value, from the crop reports download page (i.e., "the standard error of the estimate expressed as a percent of the estimate").
low_sample_size	Boolean	Flag indicating the estimate is "statistically unreliable due to a low sample size" (if set to true).

During import, we enforce a few constraints to ensure data are consistent, namely:

- There is only one instance of each measurement (i.e., unique index over year, state, crop, and variable);
- Variable values (estimate, relative_std_err, and low_sample_size) are either all defined or all null; and
- State columns (state, state_abbr, and state_fips) are either all defined or all null.

2 Census of Agriculture

The National Agricultural Statistics Service (NASS) manages and collects data for the Census of Agriculture, which is conducted once every five years. The latest census was conducted in 2017; the questionnaires were mailed to farm and ranch operators in December 2017 to collect data for the 2017 calendar year. Responses were due in February

2018 but were accepted up until early August 2018. The 2017 Census of Agriculture dataset was downloaded directly from the Census of Agriculture website.

3 COMET-Planner

The USDA Natural Resource Conservation Service (NRCS) and Colorado State University develop and manage the COMET-Planner tool. COMET-Planner provides generalized estimates of the greenhouse gas emissions and carbon sequestration benefits of a variety of conservation practices.

The COMET-Planner dataset was downloaded directly from the USDA NRCS website on March 2, 2021. The mapping of COMET_PLANNER base data to columns in UDP is detailed in the table below.

Source columns	**UDP columns**
state	STATECODE
county	COUNTYNAME
fipsint	FIPSINT
mlra	MLRA
lrr	LRR
class	CLASSNAME
cpsnum	CPSNUM
cps_name	PRACTICESTANDARD
planner_implementation	IMPLEMENTATION
n	N
lrr_fill	LRR_FILL
co2_mean	CARBONAVE
co2_sterr	CO2DEV
n2o_mean	N2OAVE
n2o_sterr	N2ODEV
ch4_mean	CH4AVE
ch4_sterr	CH4DEV
soil_carbon_co2	ASOILCARBON
soil_carbon_co2_sterr	SOIL_CARBON_CO2_STERR
biomass_co2	BIOMASS_CO2
fossil_co2	AFOSSILCO2
drainedorganicsoils_co2	DRAINEDORGANICSOILS_CO2
biomass_burning_co2	ABIOMASSBURNINGCO2
biomassburning_n2o	ABIOMASSBURNINGN2O
biomassburning_ch4	ABIOMASSBURNINGCH4
liming_co2	ALIMING
direct_soil_n2o	DIRECT_SOIL_N2O
direct_soil_n2o_sterr	DIRECT_SOIL_N2O_STERR
indirect_soil_n2o	INDIRECT_SOIL_N2O
indirect_soil_n2o_sterr	INDIRECT_SOIL_N2O_STERR
drainedorganicsoils_n2o	DRAINEDORGANICSOILS_N2O
soil_ch4	ASOILCH4
total_ghg_co2	ATER
total_ghg_co2_sterr	TOTAL_GHG_CO2_STERR
total_ghg_co2_min	AMINTER
total_ghg_co2_max	AMAXTER

4 Crop Data Management Systems (CDMS)

Crop Data Management Systems, Inc., a Proagrica company as of 2021, maintains the CDMS database of agricultural products, primarily in the U.S. The database includes crop varieties, seeds, and inputs, such as chemical applications

(e.g., fertilizers, herbicides, pesticides, etc.). The data are collected from commercial product labels.

The CDMS product dataset is ingested through CDMS's APIs and stored in Indigo's read-only datastore. CDMS data include the following views and information.

CDMS VIEW	Description
ALL_CROPS	List of detailed/ individual crops in CDMS catalog. Includes plants from categories not shown in CROPS.
BRANDS	List of seed brands: the name of the seed company (e.g., Beck's Hybrids, Asgrow, Dynagro, etc.).
CATALOG_YEARS	List of seed catalog years: the year the seed is intended for planting per the seed company. If a seed is offered in multiple years, the catalog year will represent each year the seed is available.
CROPS	List of all crops named on product labels. Crop here is the highest-level crop type to which seeds are associated (e.g., Corn, Wheat, and Soybeans).
PESTS	List of all pests named on product labels.
PRODUCTS	Table of all agricultural chemical and nutrient products, with descriptive fields. Note: Many of the fields in this table contain semi-unstructured data in the form of arrays and JSONs.
PRODUCTS_CROPS	Correspondence of PRODUCTS to CROPS.
PRODUCTS_DELETED	Table of label products that have been removed from the database.
PRODUCTS_PESTS	Correspondence of PRODUCTS to PESTS.
SEEDS	Table of all seed products, with descriptive fields. Seed Name: The name of the seed (hybrid or variety) provided by the seed brand company. Many seed names are a combination of alphanumeric characters. Relative Maturity (RM): A seed company's representation of a value (or words) for the seed to reach maturity.
SEEDS_CATALOGYEARS	Correspondence of SEEDS to CATALOG_YEARS.
SEEDS_TRAITS	Correspondence of SEEDS to TRAITS.
SUB_CROPS	Correspondence of sub-crops from the chemical/ product reference database to the Crop list (CROPS). Sub-Crop: A more specific crop type (e.g., Field Corn and Silage Corn are sub-crops of Corn, and Winter Wheat is a sub-crop of Wheat). In many cases (e.g., soybeans and cotton), there are no sub-crops, so both the Crop and Sub-Crop will be the same name.
TRAITS	List of traits: This API's use of the word "trait" is limited to biotech pest management traits. Agronomic traits such as "waxy," "high oil," and "organic" are not specified. Thus, the trait is an event within the seed. If no trait is present, the entry will be either Conventional (if specified by the seed company) or None (if not specified).

5 Cropland Data Layer (CDL)

The USDA National Agricultural Statistics Service (NASS) releases the Cropland Data Layer (CDL) annually. The data, which are released in February and reflect the crops that were harvested in the previous year's crop season, are provided as a classification with up to 250 different classes related to agricultural and non-agricultural land use. The CDL source website contains additional information about the generation and validation of the product.

Depending on the date of data entry, the latest available version of CDL (2023 or 2024) was used at a 30-m spatial resolution for the Conterminous United States (CONUS). The CDL classification data were then tiled into a standard analysis-ready grid and summarized for every crop field from our Agronomic Data Service (ADS). The output includes a tabular dataframe with the number of 30-m x 30-m pixels in each field for each class.

The dataframe includes several columns of interest. The cdl_majority column is the most likely class for that boundary according to the full histogram of counts of each class inside of that boundary. There is a total_pixels column that tells the user how many total pixels were inside that boundary. The other output columns are all like cdl_{class} where a lookup is used to link the CDL class to the Indigo Crop Types.

6 Enforcement and Compliance History Online (ECHO)

CDL datasets from 2008 to 2024 are used to confirm field eligibility. In cases where the current version of CDL was missing data, previous versions of CDL were used to fill those gaps. The U.S. Environmental Protection Agency (EPA)

manages the Enforcement and Compliance History Online (ECHO) website. ECHO is accessed directly through the website link; the tool can be used to search for facilities and assess their compliance with environmental regulations.

7 Gridded Soil Survey Geographic (gSSURGO) Database

The Gridded Soil Survey Geographic (gSSURGO) Database from the USDA National Resources Conservation Service (NRCS) contains a regridded product of soil sample data for the United States. The USDA NRCS provides additional structural documentation and metadata, including a data dictionary on their website. The latest version of gSSURGO was released in July 2020; UDP was last updated in March 2021 to reflect the July 2020 edition. The gSSURGO data included are detailed in the table below.

View	Description
CHFRAGS	The Horizon Fragments table lists the mineral and organic fragments that generally occur in the referenced horizon. If the Volume % is greater than zero (low=5, RV=10, high=15) in a row, the kind and size of fragment in that row exists everywhere this horizon and component occur in the map unit. If the Volume % includes zero (low=0, RV=5, high=10), the kind and size of fragment may exist in some places, but not in others.
CHORIZON	The Horizon table lists the horizon(s) and related data for the referenced map unit component. If the horizon thickness is greater than zero (low=5, RV=8, high=12), the horizon exists everywhere this component occurs. If the horizon thickness includes zero (low=0, RV=1, high=3), the horizon may exist in some places, but not in other places. Horizons that have two distinct parts, such as E/B or E&Bt horizons, are recorded twice. Once for the characteristics of the first part; and again on another row, using the same depths and thicknesses, for the characteristics of the other part.
COMPONENT	The Component table lists the map unit components identified in the referenced map unit, and selected properties of each component. If the Component % is greater than zero (low=65, RV=75, high=90) for a component, that component exists in every delineation of that mapunit. If the Component % includes zero (low=0, RV=50, high=90), the component may exist in some delineations, but not in others.
LAOVERLAP	The Legend Area Overlap lists the geographic areas that are coincident with the soil survey area identified in the Legend table. For example, a survey area that covers two counties would have two rows in this table, one for each county. Other types of geographic areas listed might include state, MLRA, rainfall (R) factor area, climate (C) factor area, etc. This table was added to make data aggregations easier.
LEGEND	The Legend table identifies the soil survey area that the legend is related to, and related information about that legend. This table was added to make data aggregations easier.
MAPUNIT	The Mapunit table identifies the map units included in the referenced legend. Data related to the map unit as a whole are also given.
MUPOLYGON	This table records all the soil map units of a soil survey area that are represented as one or more polygons. This table is not like other tabular data tables that are delivered as ASCII delimited files. How the information is delivered depends on the spatial format that was requested at the time the corresponding soil survey area was exported. Spatial information is typically delivered in GIS vendor specific format, and more than one output file may be produced when the information for this spatial entity is exported. Consequently, no spatially oriented columns are shown, because the name and number of columns necessary to portray spatial characteristics vary depending on spatial format. This table is documented in order to show attributes that will always be available when working with the corresponding spatial entity in a GIS. These attributes have two purposes. One purpose is to identify each instance of the corresponding spatial entity. The other purpose is to identify each instance of the corresponding tabular entity. The tabular entity corresponding to a map unit polygon is represented by a record in the Map Unit table.

View	Description
VALU1	The Value Added Look Up table is included with the gSSURGO database, but not a part of the standard SSURGO dataset. This table contains 57 pre-summarized or “ready to map” attributes derived from the official SSURGO database. These attribute data are pre-summarized to the map unit level using best-practice generalization methods intended to meet the needs of most users. The generalization methods include map unit component weighted averages and percent of the map unit meeting a given criteria. These themes were prepared to better meet the mapping needs of users of soil survey information and can be used with both SSURGO and gridded SSURGO (gSSURGO) datasets.

8 MIrAD-US

The U.S. Geological Survey (USGS) Earth Resources Observation and Science (EROS) Center develops and manages the Moderate Resolution Imaging Spectroradiometer (MODIS) Irrigated Agriculture Dataset for the United States (MIrAD-US) irrigation data products. The MIrAD-US 1km-resolution dataset was exposed and used to look up MIrAD results for a (lat, lon) point. The 1km-resolution dataset has been spatially smoothed, thereby reducing the noise inherent in sampling the dataset at a single point.

9 NASS Quick Stats

USDA National Agricultural Statistics Service (NASS) produces the Quick Stats dataset, including both “Census" and "Survey" data. Census data are based on larger samples and are thus considered more accurate; however, observations are not available for every combination of year, crop type, geography, and parameter. Online Quick Stats resources include the glossary, column definitions, ad-hoc query tool, and a visualization interface. The Quick Stats database was ingested into UDP through an API.

10 NASS Usual Planting and Harvest Dates

The National Agricultural Statistics Service (NASS) publishes the Usual Planting and Harvesting Dates handbook, which identifies the usual planting and harvesting dates by state, crop, and sub_crop for crops covered in NASS’s program of crop estimates. The handbook was compiled from a 20-year history of progress reports and supplemented with agronomic expertise. The frequency of release is not specified by the USDA. The 2010 release is the most recent; the previous edition was issued in 1997. The 2010 Usual Planting and Harvest Dates data were downloaded directly from the USDA website; the site is developed and maintained by the Mann Library at Cornell University.

11 National Wetlands Inventory (NWI)

The U.S. Fish and Wildlife Services' National Wetlands Inventory (NWI) data product is a detailed geospatial resource for wetland distribution and characteristics throughout the United States and its territories. The NWI product is continuously updated for improvements as new mapping data are collected. Updates are released on a biannual basis in May and October of each year. The provenance for such updates is included in the metadata in a few forms. The data are projected in NAD83 Albers Equal Area. For a given State or Watershed, the data included are as follows.

NWI Layer
Wetlands polygon data - Version 2
Wetlands Project Metadata - Version 2 (includes image dates and project information)
Wetlands Historic Map Information
Riparian polygon data
Riparian Project Metadata (includes image dates and project information)
Historic Wetlands
Historic Wetlands Project Metadata (includes image dates and project information)

The Wetlands polygon data layer contains polygons for all wetlands and classifies them into approximately 5,400 ATTRIBUTE categories. The Wetlands Project Metadata contains information about how the classification was produced. NWI is usually assessed based on aerial photographs, so this layer provides information including the aerial photo image source, quality, and acquisition date, as well as a few fields of comments that describe whether the data were updated and why.

The NWI is distributed as a vector dataset with only 2 non-geometry related attributes in the attribute table: ATTRIBUTE and WETLAND_TY (presumably WETLAND_TYPE). In addition to this polygon/ vector dataset, the NWI also includes an approximately 10MB CSV that can be joined onto the geometries using the ATTRIBUTE field. This dataset, the NWI Code Definitions, contains 23 additional columns that provide more detail about the ATTRIBUTE fields. The NWI documentation states that these attributes may be used to crosswalk between the U.S. Fish and Wildlife Services and NWI data (as defined by Federal Wetland Mapping Standard) to the NWI label (as defined by the Federal Wetlands Classification Standard).

12 North American Land Data Assimilation System (NLDAS)

The North American Land Data Assimilation System (NLDAS) data product is produced by the National Aeronautics and Space Administration (NASA) and published with a roughly 4-day time lag as a gridded product (GRIB). NLDAS data are ingested via a pipeline into UDP. Each NLDAS grid point is associated with a unique tile ID, which acts as a primary index for the NLDAS data set in UDP. The NLDAS data set has a resolution of 1/8th degree, or roughly 9-12 km in the U.S.

To ensure that the fields provided by NLDAS are correct and complete, we compare them against other trusted data sources, such as Automated Weather Observing System (ASOS) weather stations or Indigo Research Partners (IRP) field sensors.

13 NRCS Cover Crop Termination Guidelines

The Natural Resource Conservation Service (NRCS) and partners publish the Cover Crop Termination Guidelines to support producers in the adoption of cover crops. The Guidelines provide a pre-approved "latest end date" for termination for producers. The NRCS Cover Crop Termination Guidelines Version 4: June 2019 map was manually converted from PDF to shapefile and added to Indigo's internal reference library as a data source.

14 Protected Areas Database of the United States (PAD-US)

The United States Geological Survey Protected Area Database of the United States (PAD-US) is America's official national inventory of U.S. terrestrial and marine protected areas that are dedicated to the preservation of biological diversity and to other natural, recreation and cultural uses, managed for these purposes through legal or other effective means. PAD-US also includes the best available aggregation of federal land and marine areas provided directly by managing agencies, coordinated through the Federal Geographic Data Committee Federal Lands Working Group.

The relevant data in the PAD-US were feature classes with a "Fee" designation and "Federal" manager type. The "Fee" feature class is described as follows: Open space lands owned outright by public Federal, State, and local agencies, nonprofits, or privately owned lands.

15 Revised Universal Soil Loss Equation, Version 2 (RUSLE2)

The USDA's Natural Resources Conservation Service (NRCS) and partners developed and maintain the Revised Universal Soil Loss Equation, Version 2 (RUSLE2). The RULSE2 model, which is an improvement on Version 1, predicts rill and interrill erosion by rainfall and runoff. The model was developed to guide conservation planning, inventory erosion rates, and estimate sediment delivery.

Data on the rainfall intensity factor for water erosion (R) are detailed below.

Column Name	Description
COUNTY	The name of the county.
DATE	The date the record was added to the source database.
FIPS	The full county FIPS code (state + county code).
NAME	The name of the region. Typically the county name.
PATH	A path provided in the source database. Not used.
PRECIP_RANGE	The range of annual precipitation in inches for which the R value is appropriate (if applicable).
R_CHOICE	A descriptor (either CLIMATE_EI_CHOICE_R_RATIO or CLIMATE_EI_CHOICE_R_AND_EI_PTR) used to indicate whether an R equivalent value is present.
R_EQUIV	An R factor that takes into account erosion during snow melt.
R_FACTOR	Rainfall intensity factor for water erosion (R)
SCIENCE_DATE	The date of analysis from the source database.
STATE	The name of the state.

16 Wind Erosion Equation (WEQ)

The USDA's Natural Resources Conservation Service (NRCS) and partners developed and maintain the Wind Erosion Equation (WEQ). The WEQ estimates wind erosion impacts and soil loss; the equation and underlying model are intended to support land managers in evaluating the efficacy of conservation practices. Data on the wind erosion intensity factor (C) are detailed in the table below.

Column Name	Description
C_VALUE	Wind erosion intensity factor (C).
FNODE_	Identifier column in source data. Not used.
LENGTH	Linestring length in meters.
ROW_NUMBER	Row number from the Parquet files.
SOURCE	The data are provided in 2 separate shapefiles and are distinguished by this attribute. C <= 3 for values in that range (values not actually specified however, all records set to 3) and C for higher values.
TNODE_	Identifier column in source data. Not used.
WKT	Linestring as Well-known text

Data on the wind erosion intensity factor (C) geometries are detailed in the table below.

Column Name	Description
C_VALUE	Wind erosion intensity factor (C).
FNODE_	Identifier column in source data. Not used.
GEOMETRY	Linestring as a geometry type. Can be used directly for spatial operations.
LENGTH	Linestring length in meters.
ROW_NUMBER	Row number from the Parquet files.
SOURCE	The data are provided in 2 separate shapefiles and are distinguished by this attribute. C <= 3 for values in that range (values not actually specified however, all records set to 3) and C for higher values.
TNODE_	Identifier column in source data. Not used.
WKT	Linestring as Well-known text

17 Summary of Changes

Section	Date	Summary of Changes